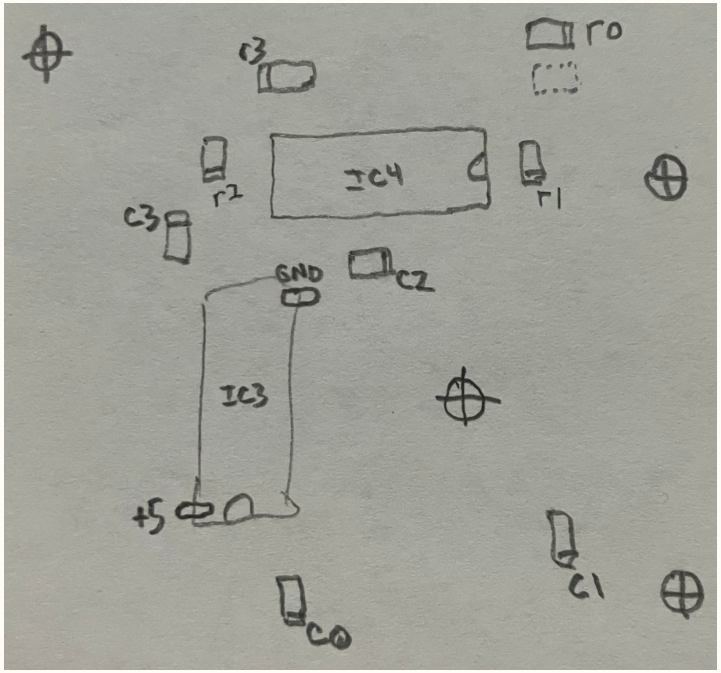


Kenwood knock-off microphone keypad-PCB parts locator for ROW, COL, and PWR connections. Remove IC3 and IC4. 1.5K resistors may remain. Connect ROW/COL wires to indicated side of each corresponding resistor pad.

TONE, MICE (tone gnd), and /MUT route to connector/MIC PCB.

If UP/DN/PTT are not needed, install R9, DNP R6 and D1.

R7 DNP 0 — 1 /UP — Q2 BSH-103 — UP 1 — GND

R8 DNP 0 — 1 /DN — Q3 BSH-103 — DN 1 — GND

+5V — R6 15.5K — D1 SBR0560S1Q-7 SOD123 — 1 /PTT

R9 DNP 0 — GND

U1 ATTINY3217-MNR

r1	ROW1	1	PA2	PA1	24	ROW0 — r0
r2	ROW2	2	ECLK/PA3	/RST/UPDI/PA0	23	
		3	GND	PC5	22	/PTT
		4	VDD	PC4	21	
r3	ROW3	5	PA4	PC3	20	COL3 — c3
		6	PA5	PC2	19	COL2 — c2
	TONE	7	PA6	PC1	18	COL1 — c1
	MMUTE	8	PA7	PC0	17	COL0 — c0
		9	PB7	PB0	16	
	UP	10	PB6	PB1	15	
	DN	11	PB5	PB2/OS2	14	TXD — R10 DNP 100 — txd
		12	PB4	PB3/OS1	13	RXD
			EP 25			

+5V — C1 1uF 0603 — GND; C2 0.1uF 0402 — GND

P1 — PGM DNP — mtg mtg — GND

/MUT — JP1 MUT_E — MICE — MGND — GND

Q1 BSH-103 — C7 0.1uF 0402 — R5 15.5K

Mic tone level is approximately 30mVrms
Need to move 4.3Vpk of DAC down to 30mVpk at center of POT rotation.

TONE — C3 1uF 0805 — R1 69.8K — R2 Nidec ST-32ETA102 1K CW — Bourns 3313X-1-103E — MGND MICE

R3 1.50K — R4 2.00K — C6 1uF 0805 — 1 TONE

C4 0.01uF 1206 — C5 1206 0.022 uF — MICE — 1 MICE

Filter Fc = 10.6 KHz, followed by 3.62 KHz

1 GND; 1 GND — GND

+5V — 1 +5V

Title	Kenwood (knock-off) DTMF Keypad Encoder	
Size: A	Number:	Revision: -
Date: 10/3/2026		Sheet of
File: C:\Users\..\KW_DTMFE.SchDoc		Drawn By: